

Module 1: Physical Layer & Signals

Network Topologies, OSI vs TCP/IP, Nyquist & Shannon Capacity, Line Coding & FDM/TDM

Module I: Transmission Fundamentals & Physical Layer Architectures

1. Data Communication Components & Network Topologies (Mesh, Star, Bus, Ring)
2. The OSI 7-Layer Reference Model vs. TCP/IP 5-Layer Protocol Architecture
3. Analog vs. Digital Signals, Composite Signals & Fourier Spectral Analysis
4. Transmission Impairments (Attenuation, Distortion, Thermal & Cross-Talk Noise)
5. Nyquist Maximum Bit Rate Formula for Noiseless Channels & Multi-Level Signaling
6. Shannon Channel Capacity Theorem for Noisy Channels & Decibel (dB) Math
7. Line Coding Schemes: NRZ-L, NRZ-I, Manchester, Differential Manchester & AMI
8. Scrambling Techniques for Synchronization: B8ZS (North America) & HDB3 (Europe)
9. Transmission Media: Guided (Twisted Pair, Coax, Fiber) vs. Unguided (Radio, Micro)
10. Optical Fiber Physics: Step-Index, Graded-Index, Single-Mode vs. Multi-Mode
11. Multiplexing Fundamentals: FDM, Synchronous TDM, Statistical TDM & WDM
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 1 & 2: Network Topologies & Layered Architectures

A **Computer Network** is an interconnected collection of autonomous computing devices capable of exchanging data frames and packets through transmission channels.

OSI Layer	Protocol Data Unit (PDU)	Core Responsibilities & Header Additions	Standard Protocols & Hardware
7. Application	Message / Data	Network virtual terminal, file transfer, directory services, user interface.	HTTP, HTTPS, DNS, SMTP, FTP, SSH
6. Presentation	Data	Data encryption/decryption (TLS), data compression (gzip), syntax translation (ASCII, UTF-8).	TLS 1.3, JPEG, MPEG, ASN.1
5. Session	Data	Dialog control, token management, session synchronization checkpoints.	RPC, NetBIOS, PPTP, Sockets
4. Transport	Segment (TCP) / Datagram (UDP)	End-to-end process-to-process delivery, port addressing, flow control, segment reassembly.	TCP, UDP, SCTP, QUIC
3. Network	Packet / Datagram	Host-to-host logical addressing (IPv4/IPv6), packet routing, fragmentation, congestion control.	IP, ICMP, OSPF, BGP (Routers)
2. Data Link	Frame	Hop-to-hop node delivery, physical MAC addressing, framing, CRC error detection, MAC access.	Ethernet (802.3), Wi-Fi (802.11), PPP (Switches)
1. Physical	Raw Bits ('0'/'1')	Transmission of uninterpreted bit streams over physical media, signal encoding, baud rates.	Manchester, NRZ, RS-232, RJ-45 (Hubs, Cables)

Topic 4 – 6: Channel Capacity Theorems & Decibel Formulations

1. Nyquist Bit Rate Formula (For Noiseless Channels):

$$\text{BitRate}_{\max} = 2 \times B \times \log_2(M) \text{ bps}$$

Where B is the analog bandwidth in Hertz (Hz), and M is the number of distinct signal voltage levels.

2. Shannon Channel Capacity Theorem (For Noisy Gaussian Channels):

$$C = B \times \log_2(1 + \text{SNR}) \text{ bps}$$

Where B is the bandwidth (Hz), and $\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}}$ is the linear Signal-to-Noise Ratio.

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}) \implies \text{SNR} = 10^{\frac{\text{SNR}_{\text{dB}}}{10}}$$

Step-by-Step Solved Problem: Shannon & Nyquist Capacity Calculation

Problem: A telephone line has a bandwidth of 3000 Hz and an $\text{SNR}_{\text{dB}} = 30 \text{ dB}$.

1. Step 1: Compute Linear SNR from dB:

$$\text{SNR} = 10^{\frac{30}{10}} = 10^3 = \mathbf{1000}$$

2. Step 2: Apply Shannon's Capacity Formula:

$$C = 3000 \times \log_2(1 + 1000) = 3000 \times \log_2(1001) \approx 3000 \times 9.967 = \mathbf{29,901 \text{ bps}} (\approx 30 \text{ kbps})$$

3. Step 3: Find Number of Signal Levels M Required by Nyquist to achieve C :

$$29901 = 2 \times 3000 \times \log_2(M) \implies \log_2(M) = \frac{29901}{6000} \approx 4.9835 \implies M = 2^{4.9835} \approx \mathbf{32 \text{ levels}}$$

Topic 7 & 8: Digital Signal Line Coding Schemes

Line Coding Scheme	Voltage Transition Rule	Key Advantages & Tradeoffs
1. NRZ-L (Level)	Bit '0' → Positive Voltage ($+V$), Bit '1' → Negative Voltage ($-V$).	Simple; suffers from DC component and baseline wander on long strings of '0's or '1's.
2. NRZ-I (Invert)	Bit '1' → Transition at start of bit interval; Bit '0' → No transition.	Self-clocking for '1's; still loses clock synchronization on continuous strings of '0's.
3. Manchester (IEEE 802.3)	Bit '0' → High-to-Low transition at bit center; Bit '1' → Low-to-High transition at center.	Zero DC component, 100% self-clocking; requires $2\times$ bandwidth ($B = 2 \times \text{data rate}$).
4. Differential Manchester	Always transitions at center. Bit '0' → Transition at start; Bit '1' → No transition at start.	Highly immune to wire polarity reversal; used in Token Ring LANs (IEEE 802.5).
5. AMI (Alternate Mark Inversion)	Bit '0' → Zero Voltage ($0V$); Bit '1' → Alternating $+V$ and $-V$.	Zero DC component; long runs of '0's lose synchronization (resolved by B8ZS/HDB3 scrambling).



Q1. State the difference between OSI and TCP/IP reference models across 5 engineering parameters. (8 Marks)

1. **Layer Count:** OSI has 7 layers (Session & Presentation separate); TCP/IP has 4 or 5 layers (combines Session, Presentation, Application).
2. **Model Origins:** OSI is a theoretical conceptual standard developed by ISO before protocols were written; TCP/IP was designed pragmatically alongside the ARPANET implementation.
3. **Service Orientation:** OSI strictly separates Service, Interface, and Protocol; TCP/IP merges services into protocols directly.
4. **Network Layer Communication:** OSI supports both Connection-Oriented and Connectionless at Network layer; TCP/IP supports only Connectionless (IP) at Network layer.
5. **Transport Layer Communication:** OSI supports only Connection-Oriented at Transport; TCP/IP supports both Connection-Oriented (TCP) and Connectionless (UDP).